Multi-Room Heat Balance

Objective

To formulate linear energy balance equations for rooms connected through walls and determine the temperatures using matrix methods and MATLAB.

Matlab Code

```
clc;
clear;
close all;

% Boundary temperatures
T_left = 100;   % degree C
T_right = 40;   % degree C

% Coefficient matrix [A]
A = [ 2  -1   0;
      1  -2   1;
      0   1  -2 ];

% RHS vector [B]
B = [T_left;
     0;
    -T_right];

% Solution using matrix (system of linear equations)
T = A\B;

disp('Temperatures at nodes:')
disp(['T1 = ', num2str(T(1)), ' deg C'])
disp(['T2 = ', num2str(T(2)), ' deg C'])
disp(['T3 = ', num2str(T(3)), ' deg C'])

% Eigenvalue analysis
[V,D] = eig(A);

disp('Eigenvalues of conduction matrix:')
disp(diag(D))

disp('Eigenvectors:')
disp(V)

% temperature distribution
x = [0 1 2 3 4];
T_plot = [T_left; T(1); T(2); T(3); T_right];

figure;
plot(x, T_plot,'-o','LineWidth',2)
grid on
xlabel('Position across rooms')
ylabel('Temperature (°C)')
title('1D Heat Conduction Temperature Distribution')
```

```
% 2D Temperature Profile
x = linspace(0,4,5);
y = linspace(0,1,60);

% Temperature distribution
T_line = [T_left T(1) T(2) T(3) T_right];

% Create 2D temperature field
[X,Y] = meshgrid(x,y);
Z = repmat(T_line,length(y),1);

% Plot 2D
figure
pcolor(X,Y,Z)
shading interp
colormap(jet)
colorbar

xlabel('Room')
ylabel('Width')
title('2D Temperature Distribution')
```

Output

```
Temperatures at nodes:
T1 = 85 deg C
T2 = 70 deg C
T3 = 55 deg C
Eigenvalues of conduction matrix:
    1.7093
   -2.9032
   -0.8061

Eigenvectors:
   -0.9575   -0.1354    0.2635
   -0.2784   -0.6641    0.7395
   -0.0750    0.7353    0.6194
```

Graph

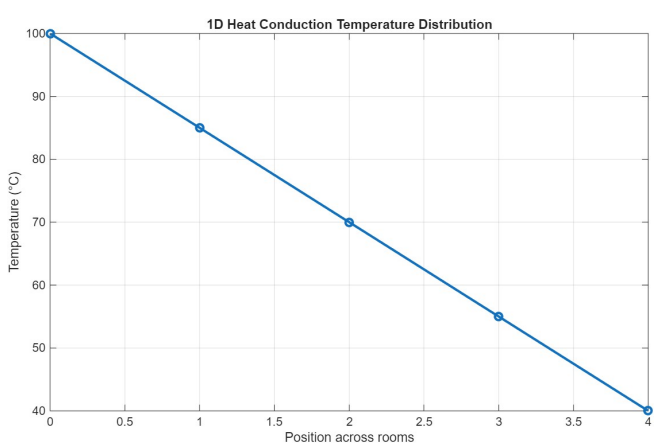

Figure 1: Variation of temperature across rooms

Figure 1 shows the steady-state temperature distribution along the rooms obtained from the matrix solution of the discretized heat conduction equations.

The temperature decreases uniformly from 100°C at the left boundary to 40°C at the right boundary. The computed interior node temperatures are:

- T1=85°C
- T2=70°C
- T3=55°C

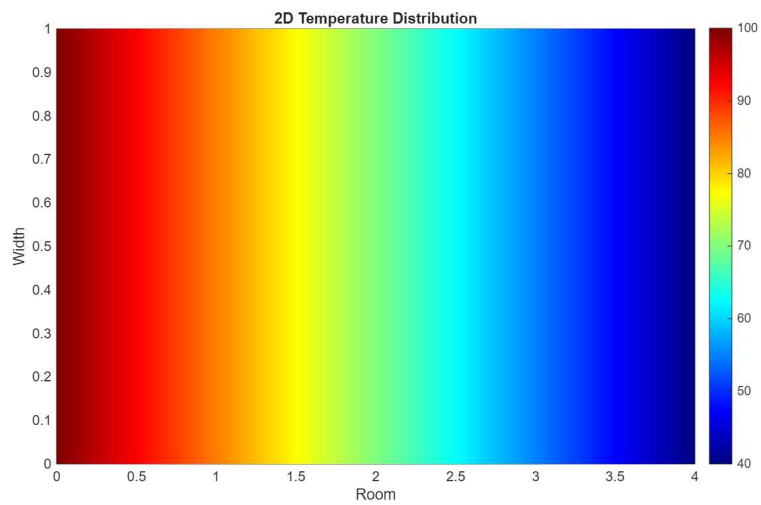

Figure 2: Temperature distribution profile

Figure 2 shows the two-dimensional temperature contour. The horizontal axis represents the sequence of rooms (or nodal positions), while the vertical axis represents the width of each room. This indicates equal thermal resistance between adjacent rooms, uniform heat transfer rate across each wall, and no internal heat generation in any room. Thus, heat flows steadily from the hottest room (Room 0) toward the coldest room (Room 4).